

ROLANDO S. GARCIA, PH.D.

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EDUCATION

University of California, Berkeley

Ph.D. in Computer Science advised by Dr. Joseph M. Hellerstein

Dissertation: *The Management of Context in the Machine Learning Lifecycle*

Berkeley, CA

2017 – 2024

Arizona State University

B.S. in Computer Science, *Summa Cum Laude*

Tempe, AZ

2014 – 2017

Northern Arizona University

Philosophy and Computer Information Systems

Flagstaff, AZ

2010 – 2013

AWARDS AND FELLOWSHIPS

ASU Presidential Postdoctoral Fellowship, two years of funding.

2024

Dell Technologies Inc. Research Award, one year of funding.

2023

NSF Graduate Research Fellowship, three years of funding.

2017

UC Berkeley Chancellor's Fellowship, two years of funding.

2017

UC Berkeley's EECS Excellence Award.

2017

CRA Outstanding Undergraduate Researcher Award, honorable mention.

2017

Fulton Undergraduate Research Initiative, research award.

2016

IEEE VAST Grand Challenge Award, outstanding comprehensive submission.

2015

RESEARCH

School of Computing & AI

Presidential Postdoctoral Fellow, advised by Dr. Ross Maciejewski

July 2024 – Present

Arizona State University

- *Submitted and under review:*

- * NSF TTP-T: *Scalable Human-in-the-Loop Review: Expert-AI Collaboration in High-Volume Document Processing*, w/ R. Maciejewski, & T. Winarski¹

- Co-PI at 40% effort, \$759k over 3 years.

- * NSF Future CoRE: *III: Explainable GNN Approaches to Ad-Hoc Queries on Heterogeneous Networks*, w/ Yixuan He² et al.

- Co-PI at 20% effort, \$1M over 3 years.

RISE Lab rise.cs.berkeley.edu

NSF Graduate Research Fellow, advised by Dr. Joe Hellerstein

June 2017 – June 2024

University of California, Berkeley

github.com/ucbrise/flor

- FlorDB: a hindsight logging database for the machine learning lifecycle.

- ... How Engineers Operationalize Machine Learning, w/ Shreya Shankar et al.

- **Internships:** Linea Labs (*startup*).

Amazon A9 (Search)

Research Intern, Search Infrastructure Team

May 2018 – December 2018

Palo Alto, CA

- Maintained model training pipelines via AWS SageMaker, Docker.

- Bulk data processing and regular re-training.

¹ Sandra Day O'Connor College of Law

² School of Mathematical and Natural Sciences

- NSF REU: Visual Analytics Algorithms for Spatiotemporal Analysis.
- Visual Analytics of Crowd Spatiotemporal Data (IEEE VAST Grand Challenge Award).

PUBLICATIONS

Cumulative Ground Truth Maintenance from Books & Documents*Submitted,*Erick Duarte, Akshit Jain, Ross Maciejewski, *Rolando Garcia*.*SIGMOD 2027**ACM SIGMOD International Conference on Management of Data*, under review.**Citation Analysis and Deterministic Evaluation at Scale***Submitted,**Rolando Garcia*, Akshit Jain, Shiyi Liu, Rostyslav Hnatyshyn, Fan Lei, Ross Maciejewski.*TIIS 2026**ACM Transactions on Interactive Intelligent Systems*, under review.**IDSPACE: A Novel Document Generator for Reliable Evaluation of Digital Identity Verification Systems***Submitted,*Lulu Xie, Yancheng Wang, Kanchan Chowdhury, *Rolando Garcia*, Yingzhen Yang, Jia Zou.*ICDM 2026**IEEE International Conference on Data Mining*, under review.**Cost-Benefit Modeling of Interactive Visual Machine Learning Systems***IEEE VIS 2026*Yixuan Wang, *Rolando Garcia*, Alex Endert, Ross Maciejewski.*IEEE Transactions on Visualization and Computer Graphics*, 2026.**Flow with FlorDB: Incremental Context Maintenance for the ML Lifecycle***CIDR 2025**Rolando Garcia*, Pragya Kallanagoudar, Chithra Anand, Sarah E. Chasins,

Erin M. Kerrison, Aditya G. Parameswaran, Joseph M. Hellerstein.

Conference on Innovative Data Systems Research, 2025.**"We Have No Idea How Models will Behave in Production until Production": How Engineers Operationalize Machine Learning***CSCW 2024*Shreya Shankar*, *Rolando Garcia**, Joseph M. Hellerstein, Aditya G. Parameswaran.*ACM Conf' on Computer Supported Cooperative Work and Social Computing 2024*.*Co-first authors. **Honorable Mention: MLOps Community Best Paper Award****Quilt: Custom UIs for Linking Unstructured Documents to Structured Dataasets***UIST Demo 2024*Pragya Kallanagoudar, Chithra Anand, *Rolando Garcia*, Rebecca M. M. Hicke,Aditya G. Parameswaran, Eunice Jun , Sarah E. Chasins. *UIST Adjunct '24:**Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, 2024.**Hindsight Logging for Model Training***VLDB 2021**Rolando Garcia*, Eric Liu, Vikram Sreekanti, Bobby Yan, Anusha Dandamudi,Joseph E. Gonzalez, Joseph M. Hellerstein, Koushik Sen. *Proceedings of the**VLDB Endowment*, 14(4): 682-693, 2021.**Context: The Missing Piece in the Machine Learning Lifecycle***KDD CMI 2018**Rolando Garcia*, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw,

Joseph E. Gonzalez, Joseph M. Hellerstein.

Workshop on Common Model Infrastructure at KDD, 2018.

A Visual Analytics Framework for Exploring Theme Park Dynamics *ACM TIIS* 2018
Michael Steptoe, Robert Krüger, *Rolando Garcia*, Xing Liang, Ross Maciejewski.
ACM Transactions on Interactive Intelligent Systems, 2018.

The State-of-the-Art in Predictive Visual Analytics *CGF* 2017
Yafeng Lu, *Rolando Garcia*, Brett Hansen, Michael Gleicher, Ross Maciejewski.
Computer Graphics Forum, 2017.

TALKS

Incremental Context Maintenance for the ML Lifecycle *CIDR* 2025
Rolando Garcia, Pragya Kallanagoudar, Chithra Anand, Sarah E. Chasins,
Erin M. Kerrison, Aditya G. Parameswaran, Joseph M. Hellerstein.
Conference on Innovative Data Systems Research. January 2025.
Amsterdam

How Engineers Operationalize Machine Learning *CSCW* 2024
Shreya Shankar, *Rolando Garcia*, Joseph M. Hellerstein, Aditya G. Parameswaran.
ACM Conf' on Computer Supported Cooperative Work and Social Computing.
November 2024. Costa Rica

Flow with FlorDB: Agile MLOps without Compromises *Mines Seminar*, 2024
Rolando Garcia. Colorado School of Mines. October 2024. Golden, CO

Operationalizing Machine Learning: An Interview Study *Flyte*, 2022
Rolando Garcia, Shreya Shankar. December 2022.

Operationalizing Machine Learning: An Interview Study *Inaugural Address*, 2022
Shreya Shankar, *Rolando Garcia*. Berkeley, CA, October 2022. EPIC, UC Berkeley

Operationalizing Machine Learning: An Interview Study *Canva*, 2022
Shreya Shankar, *Rolando Garcia*. October 2022.

Operationalizing Machine Learning: An Interview Study *MLOps World*, 2022
Shreya Shankar, *Rolando Garcia*. San Francisco Bay Area, October 2022. San Francisco

FLOR: Managing the Infinite Virtual Metadata of Model Training *Cornell Seminar*, 2022
Rolando Garcia. Cornell University, February 2022.

Hindsight Logging for Model Training *VLDB* 2021
Rolando Garcia, Eric Liu, Vikram Sreekanti, Anusha Dandamudi, Bobby Yan,
Joseph Gonzalez, Koushik Sen, Joseph Hellerstein. *International Conference*
on Very Large Databases. August 2021. Copenhagen

Choose What to Log, After you Execute *RISE Retreat* 2019
Rolando Garcia, Eric Liu, Vikram Sreekanti, Anusha Dandamudi, Bobby Yan,
Joseph Gonzalez, Koushik Sen, Joseph Hellerstein. Retreat talk to lab sponsors.
Monterey, CA. January 2019.

ML in Context: Preserve, Share, and Analyze Experiments with Flor *RISE Camp* 2018
Rolando Garcia, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw,
Sona Jeswani, Eric Liu, Malhar Patel, Joseph Gonzalez, Joseph Hellerstein.
Talk and Tutorial for lab sponsors. Berkeley, CA. October 2018.

Context: The Missing Piece in the Machine Learning Lifecycle <i>Rolando Garcia</i> , Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw, Joseph Gonzalez, Joseph Hellerstein. <i>The 24th ACM SIGKDD Intl' Conf.</i> August 2018.	KDD 2018 London
Data Management Challenges in the Machine Learning Lifecycle <i>Rolando Garcia</i> , Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Sona Jeswani, Eric Liu, Malhar Patel, Joseph Gonzalez, Joseph Hellerstein. Invited talk at Amazon Search. Palo Alto, CA. April 2018.	Amazon 2018 Palo Alto
Design, Run, and Reproduce Experiments with Jarvis <i>Rolando Garcia</i> , Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Alexey Tumanov, Joseph Gonzalez, Joseph Hellerstein. Retreat talk to lab sponsors. Monterey, CA. January 2018.	RISE Retreat 2018
Jarvis: Managing the Machine Learning Lifecycle <i>Rolando Garcia</i> , Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Alexey Tumanov, Joseph Gonzalez, Joseph Hellerstein. Invited talk at Google. Mountain View, CA. September 2017.	Google 2017 Mountain View
The State-of-the-Art in Predictive Visual Analytics Yafeng Lu, <i>Rolando Garcia</i> , Brett Hansen, Michael Gleicher, Ross Maciejewski. 19 th EG/VGTC Conference on Visualization, Barcelona, Spain. June 2017.	EuroVis 2017 Barcelona
Automatic Design and Optimization of Deep Learning Models <i>Rolando Garcia</i> , Randal Olson, Jason Moore. <i>The Leadership Alliance National Symposium</i> , Stamford, CT. July 2016.	LANS 2016
VAST 2015 Grand Challenge – Team VADER/VIS Michael Steptoe, Robert Krüger, Yifan Zhang, Xing Liang, <i>Rolando Garcia</i> , Sagarika Kadambi, Wei Luo, Thomas Ertl, Ross Maciejewski. <i>IEEE Visual Analytics Science and Technology Conference</i> , Chicago, IL. October 2015.	VAST 2015 Chicago

TEACHING

Introduction to Database Systems (COMPSCI 186/286) http://cs186berkeley.net Graduate Student Instructor, 450 students. Platform: bCourses, Blackboard, Gradescope, Piazza	Spring 2018 <i>University of California, Berkeley</i>
Introduction to Programming Languages (CSE 240) Undergraduate Student Instructor, 267 students	Fall 2015 <i>Arizona State University, Tempe</i>
Introduction to Ethics (PHI 105) Undergraduate Student Instructor, 50 students	Fall 2013 <i>Northern Arizona University, Flagstaff</i>

SERVICE

Diversifying Access to Research in Engineering (DARE) Organizer dare.berkeley.edu	June 2020 – December 2022 <i>University of California, Berkeley</i>
EECS Graduate Admissions Graduate Student Representative eecs.berkeley.edu	Fall 2021 <i>University of California, Berkeley</i>
Data Systems and Foundations Seminar Coordinator dsf.berkeley.edu	Fall 2020 <i>University of California, Berkeley</i>

UC Berkeley / Stanford Graduate Pathways to STEM
Peer Advising Co-Chair and Peer Advisor | [GPS Hyperlink](#)

June 2017 – December 2019
UC Berkeley & Stanford